

VINDSOL

Harnessing the nature's energy for you

Technical Data Sheet - Domestic Thermodynamics Solar Heat Pump

mini series

india

World class technology
Proudly designed and made in India



NEXT-GEN THERMODYNAMICS SOLAR HEAT PUMP

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The Thermodynamic Solar System joins two incomplete technologies, the heat pump and the solar thermal collector. Heat pumps are quite efficient equipment but the heat they produce from their renewable component varies only according to changes in the temperature of the environment.

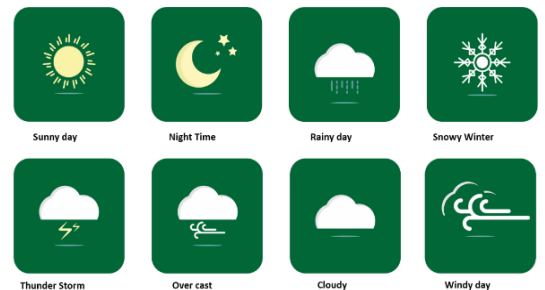
Thermal solar collectors are the best source of heat on hot and sunny days, but they are totally inefficient whenever there is no sun.

The thermodynamic Solar Technology manages to surpass the limitations of both the heat pump and solar collector. Through the cooling liquid (R134a or R407c) which covers a closed circuit, the liquid goes into the solar panel and extract the thermal energy of sun, rain, wind, environment temperature and other climate factors. During this process the liquid gains heat in a more favorable way than a heat pump. After this stage, the heat is transferred to an exchanger with the help of a small compressor, which heats the water. The liquid cools down and the circuit is repeated.

As the fluid has a boiling temperature of approximately -30°C , the system works even when there is no sun and it even works at night, hail, rain, wind or shine, unlike the traditional solar thermal system.

The energy consumption of the system is basically the same as fridge compressor that makes the liquid circulate. There are no ventilators that help the evaporation process, of defrost cycles, which imply unnecessary energy consumption, unlike what happened with heat pump.

The Thermodynamic Solar technology offers high performance even at night, with rain, or on overcast days, this being an advantage over conventional solar panels whose performance is limited as they rely exclusively on direct solar radiation and a compulsory southward orientation (Northern Hemisphere).



NEXT-GEN THERMODYNAMICS SOLAR PANEL COLLECTOR

Latest generation thermodynamic solar collector: it is a roll-bond panel with double channel through which the fluid refrigerant circulates, being able of providing high performance at night and in adverse climate conditions.

Life span of more than 10 years: the panel is made of anodized aluminum 30 micron, which ensures

The following properties

- Great toughness at high and low temperatures.
- High surface hardness.
- Resistance to abrasion and weakening.
- High protection against dust and dampness.
- Corrosion-resistant.

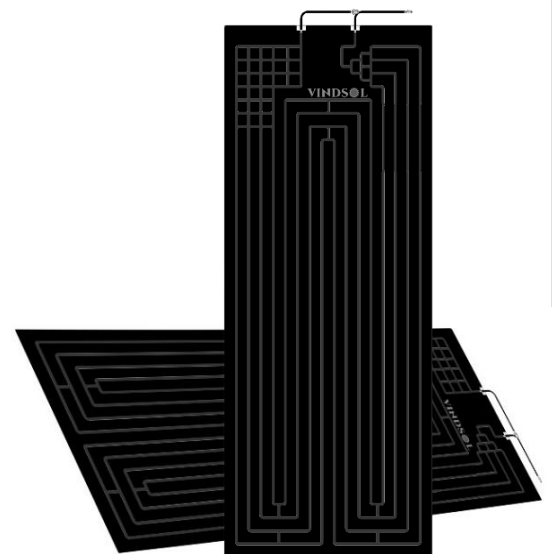
User-friendly transport and easy handling systems: small size 2 m x 0,8 m.

Versatility: easy integration on facades, roofs or any other surface. It is not necessary to reinforce structures as it is a lightweight panel, 8 kg.

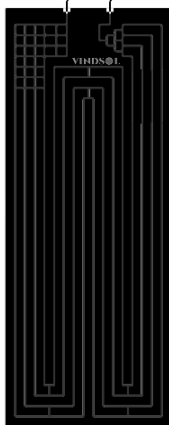
Optimized solar catchment area: both sides of the panel, 3,20 sqm.

No need to be south orientated (Northern Hemisphere). Vindsol recommends to orientate the collector as much as possible to the sun and exposed to the elements, such as wind and rain.

Regular maintenance is not necessary.



NEXT-GEN THERMODYNAMICS SOLAR HEAT PUMP

TECHNICAL SPECIFICATION				
Model		MiNi4-500		
Thermodynamic Solar Panel				
Material & Finish		Powder coated Anodised aluminium Alloy panel		
Thickness	L X W X T (mm)	2 mm aluminium for increased mechanical strength		
Protection		Corrosion resistant and adaptability to all weathers		
Dimensions L X W X T	mm	2000 X 800 * 20		
Maximum working Pr.	Bar	10		
Test pressure	Bar	15		
Maximum exposure temp.	°C	100		
Minimum running temp.	°C	10		
Qty	Nos	8		
Thermodynamic Box				
Absorbed power KW (Average/Max)		1.87		
Thermal Calorific power (Average/Max)		4750		
Compressor Type		Rotary Type - Heat pump COmpressor		
Throttle Type		Electronic expansion valve (EEV)		
Rated outlet water temp.°C		50-55 Deg		
Max outlet water temp.°C		55 Deg		
Power V/Ph/Hz		220V/1PH/ 50Hz		
Heat exchanger type		High efficiency tube in tube heat exchanger		
Heat exchanger No.		1		
Controller type		Intelligent PCB & LCD		
Ambient Temperature		(0°C~+45 °C)		
Low pressure protection		Yes		
High pressure protection		Yes		
Automatic defroting system		Yes		
Overload protection		Yes		
Noise in 1 meter		50 dB		
Water pipe size		25 NB		
Water flow volume		2 M3/hr		
Circulation Pump		Additional		
Net Weight	Kg	50 Kg Apprx		
Cabinet	Type	Stainless Steel / Galvanised Steek		
Dimension	LxWXH	700 X 700 X 1000		
Electrical support power		2 Kw		
Circulation Pump		Bultin Hot water circualtor Pump		
Cabinet	Type	Stainless Steel / Galvanised Steek		
Dimension	LxWXH	500 X 450 X 375		

Refrigerant		
Cooling Fluid *	R134a	
Piping Material	Copper (DHP ISO1337)	
Line of liquid	3/8"	
Line of aspiration	1/2"	

Storage Water Heater			
Volume	Litres	500 Lts Imported Glass Lined	
Type of internal Tank		Cellipsoidal RC premium Steel	
Outer Shell material		Pre Coated GI 0.5 mm	
Inner Tank Dia		610 mm	
Outer tank dia		710 mm	
Coating of internal Tank		QuartzBlue Glass lining	
Tank Insualtion		High density compressed PUF insualtion 50 mm	
Water Inlet / Outlet	In	3/4"	
Circulation water inlet/outlet	In	3/4"	
Working pressure	Bar	7	
Test pressure	Bar	10	
Maximum water temp.	°C	75	
Water temperature	°C	55	

ELECTRIC BOARD - ONE TOUCH PCB CONTROLLER

One touch intelligent controller with Touch screen design, clear working condition displaying, and self-check function of error.		
Power feed	220V/1PH/ 50Hz	
Controller	Next Gen PCB controller	

Testing condition : * Heating: Ambient temp. (DB/WB): 20°C/15°C, water temp. (input/output): 15°C/55°C. The above data is only for reference: specific data is subject to the product

COPPER TUBING aand plumbing UNITS AND ACCESSORIES



COPPER TUBE WITH INSULATION



CONNECTION FITTINGS FOR TANK AND HEAT PUMP

“Consumables such as copper piping, including necessary accessories, fittings, and insulation, are considered for a standard length of 5 meters. Any additional requirement beyond this will be under the customer’s scope, depending on site conditions. The complete plumbing work, including insulation for the distribution line, shall also fall within the customer’s scope.”

Commercial Offer

		Unit	Qty	Rate	Amount
Vindsol Make Next Gen Commercial Thermodynamics solar heat pump for 500 lts hot water capacity , complete with heat pump + 4 nos Thermodynamics Panel + 500 Lts Storage Tank + fixing clamp Accessories		Set	1	₹ 2,98,000	₹ 2,98,000
Vindsol Make Flat plate solar water collector of 125 LPD capacity Copper - Copper fin and tubes along with stand and frame		No	4	₹ 19,500	₹ 78,000

Safety Kit Pack for Pressurised Tank - Additional Required

	Vacuum Relief Valve GST @ 18% extra	WATTS imported 3/4 inch lead free Vacuum Relief Valve automatically allows air to enter into the piping system to prevent vacuum conditions that could siphon water from the system and damage water equipment.	included	1	
	Safety Pressure Relief Valve GST @ 18% extra	Safety Pressure Relief Valve 1/2" 3 Bar , Protects hot water cylinders and heating components from high pressure			
	Automatic Air Vent valve GST @ 18% extra	Brass Automatic Air Vent valve 1/2" BSP male thread. It automatically removes trapped air from a central heating installation.			
	Water Circulation pumps GST @ 18% extra	Mechlion Make Hot water circulation pump between Heat pump and tank multistage pump Model : HWS 25-7	included	1	
			Total		₹ 3,76,000

TERMS & CONDITIONS

Prices applicable are Ex.Works , Bangalore

Packing and forwarding cost 2 % extra

Payment to be made 100% prior to delivery

Delivery : ex stock else within 4-5 weeks

Panel Installtion Structure will be in customer scope

Online Installation and commissioning Support will be provided

Note: “Standard wall-mounting clamps are supplied along with the thermodynamic panels. Any additional or site-specific installation requirements, including special orientation or structural arrangements, shall be the responsibility of the customer and must be carried out in accordance with our instructions.”

VINDSOL

Complete water Heating Solutions

Mechzephyr Engineering Pvt Ltd

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Commercial Water Heat Pumps
Next Gen Thermodynamics Heat Pumps
Swimming pool heat pump